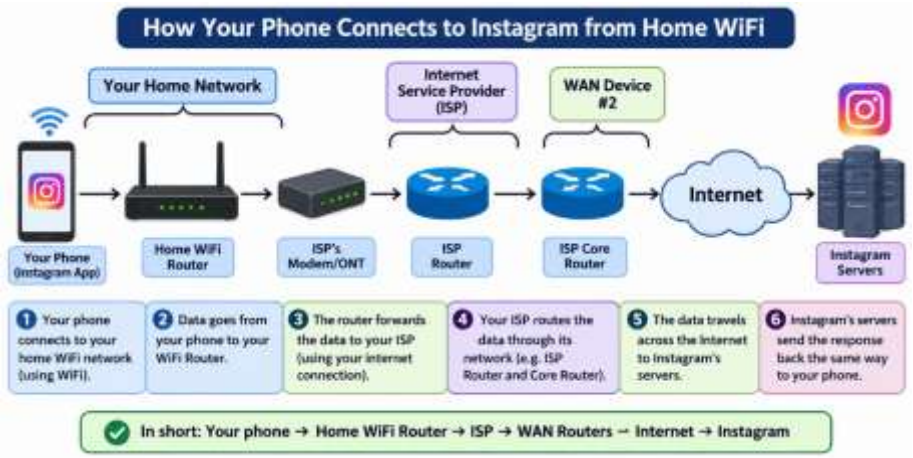
Wide Area Networks

Task 1: Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

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1. Network Diagram

The following diagram shows how my mobile phone connects to Instagram's servers using my home Wi-Fi network and the Internet Service Provider (ISP).



2. Devices Used in the Network

1. Mobile Phone: Runs the Instagram application and sends data.
2. Home Wi-Fi Router: Connects the phone to the home network wirelessly.
3. ISP Modem/ONT: Connects the home router to the Internet Service Provider.
4. ISP Router: Forwards data through the ISP network.
5. ISP Core Router: Routes data across the ISP's Wide Area Network (WAN).
6. Instagram Servers: Receive requests and send Instagram content back to the phone.

3. Step-by-Step Working

1. I open the Instagram application on my mobile phone.
2. My phone connects to the home Wi-Fi router using a wireless connection.
3. The Wi-Fi router forwards the data to the ISP modem/ONT.
4. The ISP router and ISP core router forward the data through the WAN.
5. The data travels through the Internet and reaches Instagram's servers.
6. Instagram's servers process the request and send the response back through the same network path to my phone.

4. WAN Devices Used

- ISP Router: Routes data between the ISP network and other networks.
- ISP Core Router: Transfers data across large geographical areas through the ISP's WAN.

5. Conclusion

This diagram explains how my mobile phone connects to Instagram's servers through a home Wi-Fi network. The ISP and WAN routers help transfer data over long distances. This allows me to access Instagram posts, reels, messages, and other content using the Internet.

Task 2: Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

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1. Introduction

WAN stands for Wide Area Network. It is a computer network that connects multiple computers, offices, and networks over large geographical distances.

WAN technologies are used by businesses, banks, government organizations, and homes to communicate and share data over long distances.

There are three common types of WAN technologies:

1. MPLS (Multiprotocol Label Switching)
2. Leased Line
3. Broadband Internet

2. Comparison Table of WAN Technologies

WAN Technology	Speed	Cost	Real-Life Usage
MPLS	High speed and reliable performance	High	Zomato offices connecting different branch offices securely.
Leased Line	High speed with dedicated bandwidth	Very High	IRCTC ticketing systems and banks requiring reliable connectivity.
Broadband Internet	Medium to high speed, depending on the plan	Low	Home Wi-Fi, small offices, and internet browsing.

3. Explanation of Each WAN Technology

1. MPLS (Multiprotocol Label Switching)

- MPLS is a WAN technology used to connect multiple office locations.
- It provides reliable network communication and traffic management.
- It is commonly used by large companies and organizations.
- Example: Zomato can use MPLS to connect its offices in different cities.

2. Leased Line

- A leased line is a dedicated communication connection between two locations.
- It provides consistent bandwidth and reliable connectivity.
- It is generally more expensive than broadband.
- Example: Banks and large organizations may use leased lines for critical business operations.

3. Broadband Internet

- Broadband is a high-speed Internet connection provided by an ISP.
- It is commonly available through fiber, cable, or DSL.
- It is affordable and suitable for home and small office use.
- Example: Using JioFiber or Airtel Xstream Fiber at home for browsing, online classes, and video streaming.

4. Conclusion

MPLS, Leased Line, and Broadband Internet are three important WAN technologies. MPLS is useful for connecting multiple business locations, leased lines provide dedicated and reliable connectivity, and broadband is an affordable option for homes and small offices.

Task 3 **Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.

Hint: Focus on how the VPN creates a private tunnel over the public WAN.**

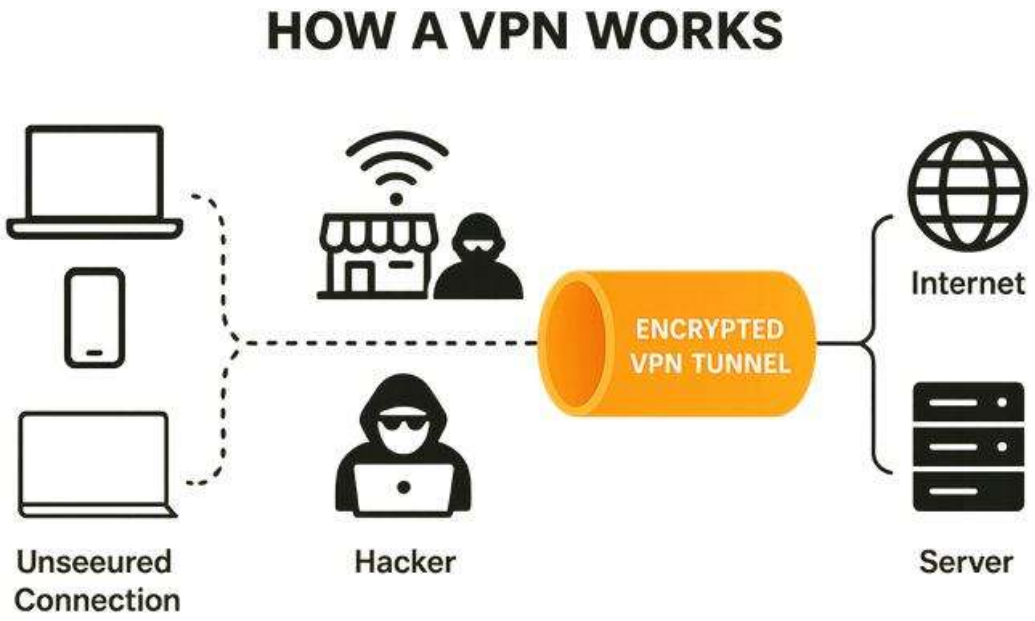
1. Introduction

VPN stands for Virtual Private Network. It is a technology that creates a secure and encrypted connection over a public network, such as public Wi-Fi or the Internet.

A VPN uses a Wide Area Network (WAN), such as the Internet, to securely transfer data between a user's device and a VPN server.

For example, when I visit a coffee shop and connect my mobile phone to public Wi-Fi, I can use a VPN to access applications like Spotify, YouTube, Instagram, and other websites more securely.

2. How VPN Works Over a WAN



The VPN works through the following steps:

1. Connect to Public Wi-Fi: I connect my mobile phone or laptop to the coffee shop's public Wi-Fi network.

2. Start the VPN: I open a VPN application and connect to a VPN server.

3. Create a Secure Tunnel: The VPN creates an encrypted private tunnel over the public WAN (Internet).

4. Encrypt Data: My data is encrypted before it travels through the public Wi-Fi network. This helps prevent others on the network from reading my data.

5. VPN Server: The encrypted data reaches the VPN server, where it is decrypted and forwarded to the requested application server.

6. Access Applications: I can securely access applications like Spotify and YouTube through the VPN connection.

7. Return Data: The response travels back through the VPN tunnel to my device.

3. Real-Life Example

Suppose I visit a coffee shop and connect my laptop to its free public Wi-Fi.

- I open my VPN application and connect to a VPN server.
- I open YouTube and play a video.
- My Internet traffic travels through an encrypted VPN tunnel over the public WAN.
- The VPN server forwards my request to YouTube.
- YouTube sends the video data back through the VPN connection.

This helps protect my Internet traffic from being viewed by other users on the public Wi-Fi network.

4. Advantages of Using a VPN

1. Security: Encrypts Internet traffic between my device and the VPN server.
2. Privacy: Helps hide my browsing traffic from other users on public Wi-Fi.
3. Data Protection: Reduces the risk of data interception on untrusted networks.
4. Remote Access: Allows employees to securely access their company networks from different locations.

5. Conclusion

A VPN uses a public WAN, such as the Internet, to create a private and encrypted tunnel between a user's device and a VPN server. It protects data while travelling over public Wi-Fi and provides a more secure way to access applications like Spotify, YouTube, and other online services.

Result: Successfully understood how a VPN uses WAN technology and an encrypted private tunnel to provide secure Internet access over public Wi-Fi.

Task 4 magine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

1. Introduction

Flipkart wants to connect its three warehouses located in Ahmedabad, Mumbai, and Delhi. For this purpose, it needs a reliable and secure Wide Area Network (WAN) technology to communicate and share data between all warehouse locations.

2. Recommended WAN Technology

Recommended: MPLS

I would recommend MPLS (Multiprotocol Label Switching) for connecting Flipkart's warehouses.

3. Two Reasons for Choosing MPLS

1. Reliable and Fast Connectivity

MPLS provides reliable network connectivity and supports efficient data transfer between different locations. It helps Flipkart synchronize inventory, orders, and warehouse information across Ahmedabad, Mumbai, and Delhi.

2. Secure and Efficient Communication

MPLS supports private network connectivity and traffic management. It helps Flipkart manage communication between warehouses efficiently and prioritize important business applications.

4. Real-Life Example

Suppose a customer orders a product from Flipkart in Delhi, but the product is available in the Ahmedabad warehouse.

- The Delhi warehouse can communicate with the Ahmedabad warehouse.
- The warehouses can share inventory and order information.
- The company can track product availability and delivery updates.
- All three warehouses can communicate through the MPLS network.

5. Conclusion

MPLS is the best WAN technology for Flipkart because it provides reliable connectivity, efficient communication, and traffic management between geographically separated warehouses.